NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

OCCUPATIONAL LUNG DISEASES: PNEUMOCONIOSES

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Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

- Describe and discuss the pathogenesis of **asbestos** related lung disease as it relates to diffuse interstitial fibrosis and correlate the clinical and pathological (gross and microscopic) features
- Briefly describe the development of benign and malignant pleural lesions in relation to asbestos exposure
- Describe the clinical and pathological features of **berylliosis** and discuss its immune pathogenesis
- Discuss, describe, and explain the pathogenesis and various forms of **coal workers pneumoconiosis** as well as the clinical, microscopic, and gross features
- Discuss, describe, and explain the pathogenesis and histological profile of **silicosis** with clinical correlation and delineation of various manifestations

Derives from the Greek
pneuma = air
konis = dust

Non-Neoplastic lung reaction to inhalation of mineral dusts encountered in workplace, also includes disease induced by chemical fumes and vapors

PNEUMOCONIOSIS

Development depends on

- Amount in lungs
- Size, shape, buoyancy
- Solubility and physiochemical reactivity
- Additional irritants (smoking)

Most dangerous 1 – 5 um

Macrophages ingest particles and immune system is stimulated

Robbins and Cotran, Pathological Basis of Disease, 11th ed., 2026, Ch. 15

Asbestos - Background

- Asbestosis - pneumoconiosis caused by inhalation of asbestos fibers.
- Characterized by slowly progressive, diffuse pulmonary fibrosis
- Pulmonary disorders associated with asbestos exposure include:
  - Asbestosis (Interstitial fibrosis)
  - Pleural disease (focal and diffuse benign pleural plaques), pleural effusions
  - Malignancies (non-small cell and small cell carcinoma of the lung as well as malignant mesothelioma)

Robbins and Cotran, Pathological Basis of Disease, 11th ed., 2026, Ch. 15

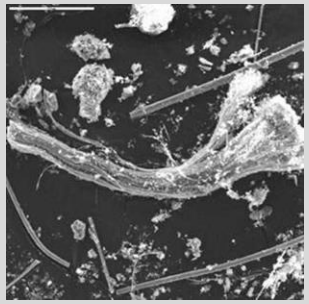
Asbestos - Background

Naturally occurring fibers - hydrated magnesium silicates

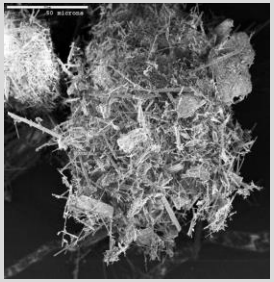
- Tensile strength, chemical properties suited for construction and insulation.
- **Asbestos fibers - two categories based on shape**
  - **Serpentine fibers** - long, curly strands (eg. chrysotile) - accounts for over 90 percent of the asbestos in US

    -**less toxic** than the amphibole fibers

    -Because of structure (curve), serpentine fibers more likely to get impacted in upper respiratory tract and **removed by mucociliary elevator**



  - **Amphibole fibers** - long, straight, rod-like structures. (eg. crocidolite, amosite, tremolite, and others)
    - **more pathogenic** especially for induction of mesothelioma, because of aerodynamic properties and solubility



Robbins and Cotran, Pathological Basis of Disease, 10th ed., 2020, Ch. 15

Asbestos - Exposure

- Mining and milling of the fibers
- Industrial applications of asbestos: textiles, cement, friction materials, insulation, shipbuilding
- Roofing, plumbing
- Renovation or demolition of asbestos-containing buildings
- Environmental exposure in the neighborhood of industrial sources, and natural environmental exposure to geological sources

Asbestos - Pathogenesis

- Direct **toxic effects of fibers** on pulmonary parenchymal cells
- Release of various **mediators (reactive oxygen species, proteases, cytokines, and growth factors)** by inflammatory cells
- **Chronic deposition of fibers and persistent release of mediators lead to interstitial pulmonary inflammation and interstitial fibrosis**
- Deleterious reactive oxygen and nitrogen free radicals may be formed either via reactions catalyzed by iron molecules within asbestos fibers or because of the activation of inflammatory cells
- Free radicals can react with and damage cellular macromolecules disrupt DNA to give rise to malignancy

Robbins and Cotran, Pathological Basis of Disease, 11th ed., 2026

Asbestos - Pathogenesis

Smoking increases the attack and/or progression rate of asbestosis, probably by interfering with mucociliary clearance of inhaled fibers

- 5 x increase in lung cancer (bronchogenic carcinoma) when exposed to asbestos
- 55 x increase in lung cancer (bronchogenic carcinoma) when exposed to asbestos and is a smoker
- Increased incidence of lung cancer **(bronchogenic carcinoma > mesothelioma)**

Robbins and Cotran, Pathological Basis of Disease, 11th ed., 2026

Asbestos - Clinical

- Most patients asymptomatic for at least 20 to 30 years after the initial exposure
- Insidious onset of breathlessness with exertion (dyspnea)
- Progresses even in absence of further asbestos exposure

Robbins and Cotran, Pathological Basis of Disease, 11th ed., 2026

Asbestos - Clinical

- Bibasilar, fine end-inspiratory crackles
- Clubbing
- Cor pulmonale
- Peripheral edema
- Jugular venous distension
- Hepatojugular reflux
- Right ventricular heave or gallop
- May lead to respiratory failure or death







Cough, sputum production, and wheezing are unusual – if present, tend to be because of cigarette smoking rather than asbestos-induced lung disease

More advanced disease can be associated with hypoxemia at rest

Asbestos - Labs

Bloodwork is generally nonspecific and not useful

Pulmonary function tests:

- Reduced lung volumes, particularly the vital capacity and total lung capacity
- Decreased pulmonary compliance
- Absence of airflow obstruction by spirometry (normal ratio of the forced expiratory volume in one second to forced vital capacity)

Robbins and Cotran, Pathological Basis of Disease, 11th ed., 2026, Ch. 15

Asbestos – Imaging

Chest Xray:

- Small bilateral parenchymal opacities with multinodular or reticular pattern, often with associated pleural abnormalities
- Interstitial process typically begins in lower lung zones
- Bilateral mid-lung zone plaques on parietal pleura
- Honeycombing and upper lobe involvement develop in advanced stages of disease
- Hilar and mediastinal lymphadenopathy are **not seen with asbestosis** and should suggest presence of another process.

Robbins and Cotran, Pathological Basis of Disease, 11th ed., 2026, Ch. 15

Asbestos – Imaging

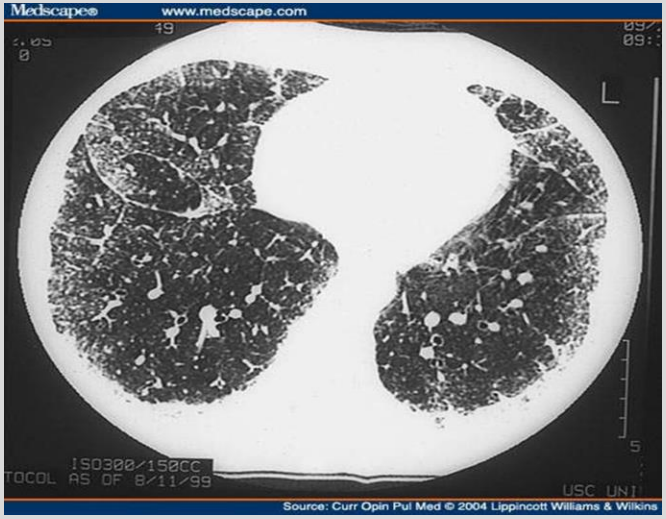
- Pleural involvement - hallmark of asbestos exposure
- Benign exudative pleural effusions
- Occur within 15 years of first exposure
- May resolve spontaneously, but leave visible blunting of the costophrenic angle or thickening of the visceral pleura





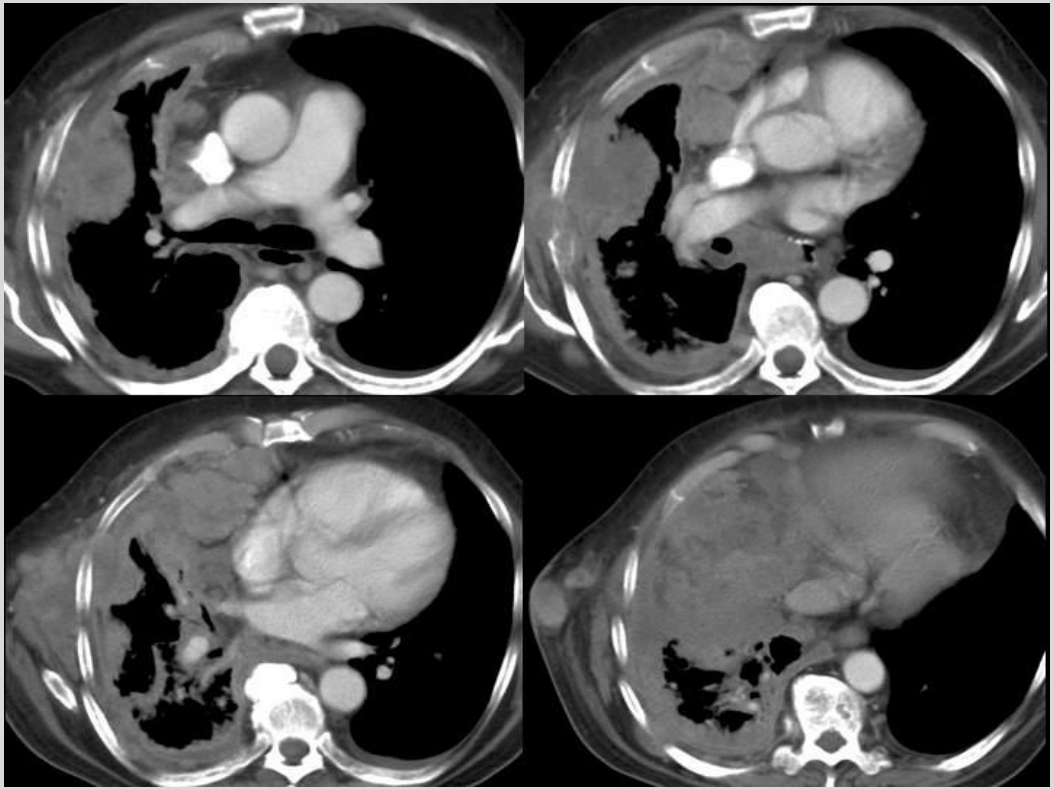
Asbestos – Imaging

Honeycombing





Malignancy



Asbestos – Further Testing

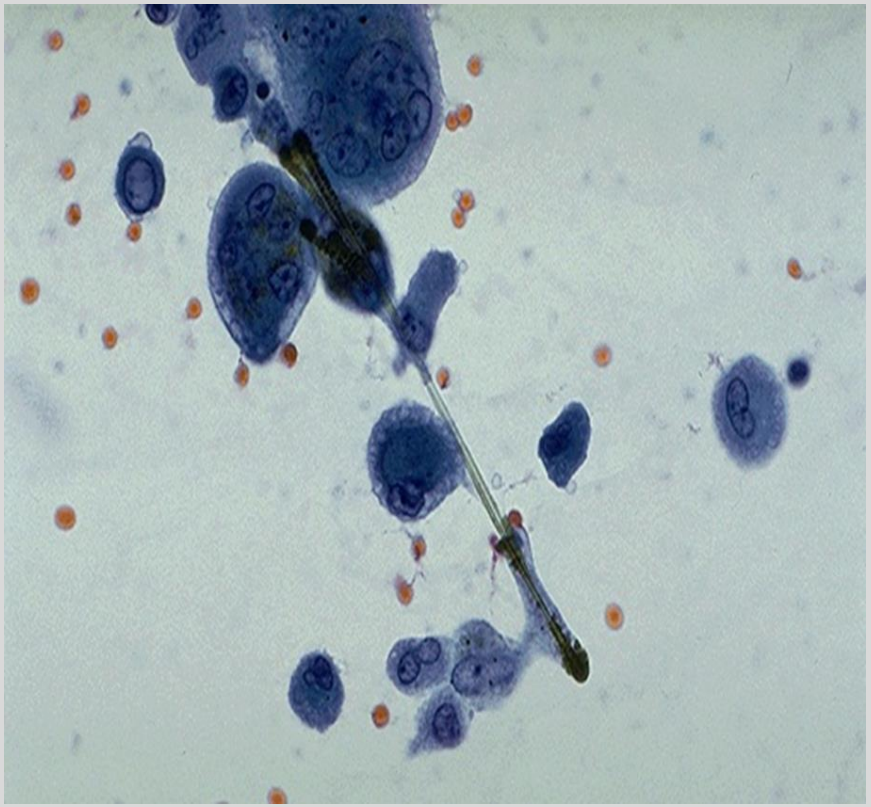
Golden brown, fusiform or beaded rods with translucent center made of asbestos fibers coated with **iron-containing** proteinaceous material.

Arise when macrophages (alveolar and interstitial) attempt to phagocytose asbestos fibers

Once fibers are phagocytosed by macrophages **release of proinflammatory factors along with fibrogenic mediators is stimulated**

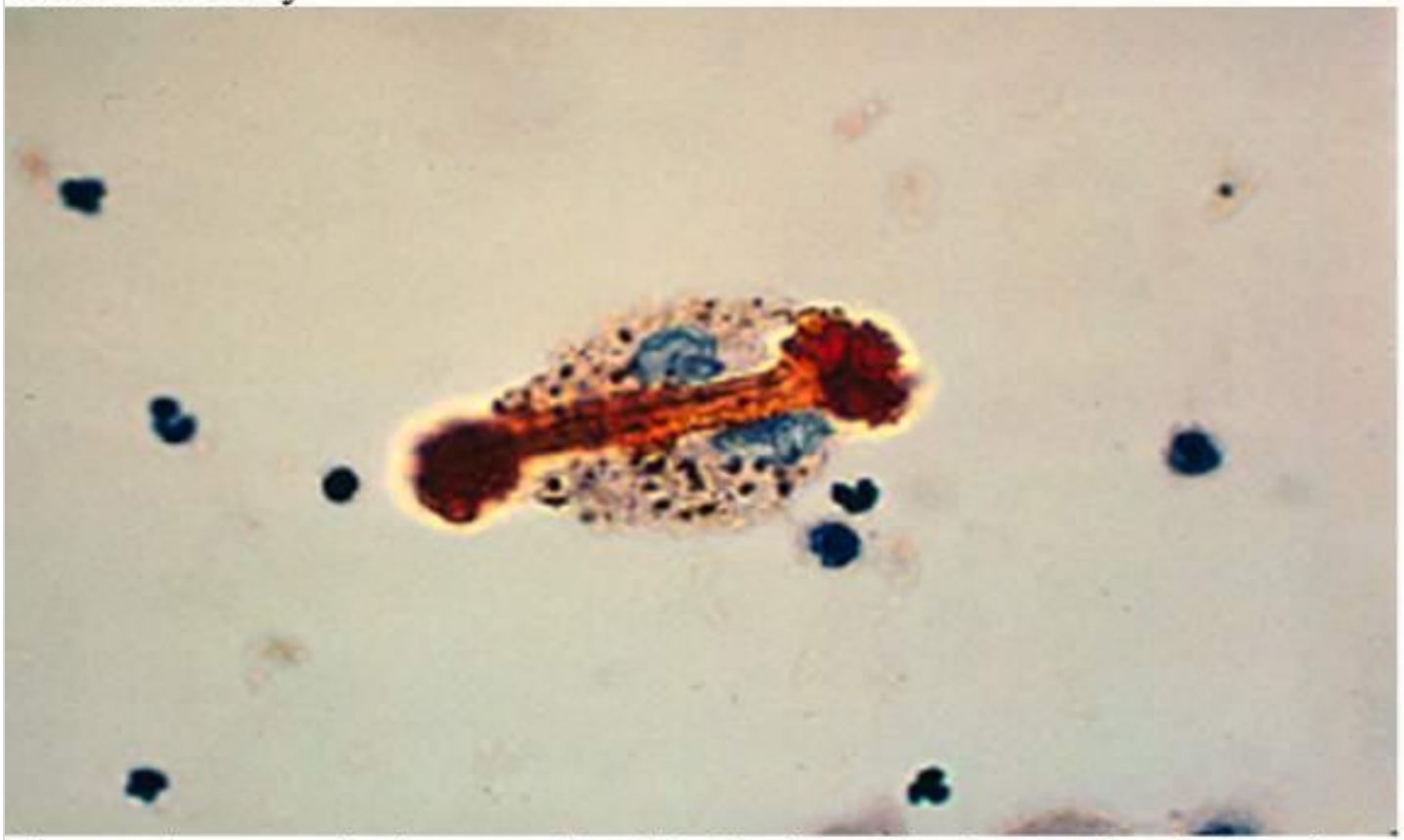
Iron assumed to come from phagocyte ferritin

****Ferruginous body** = **ASBESTOS****



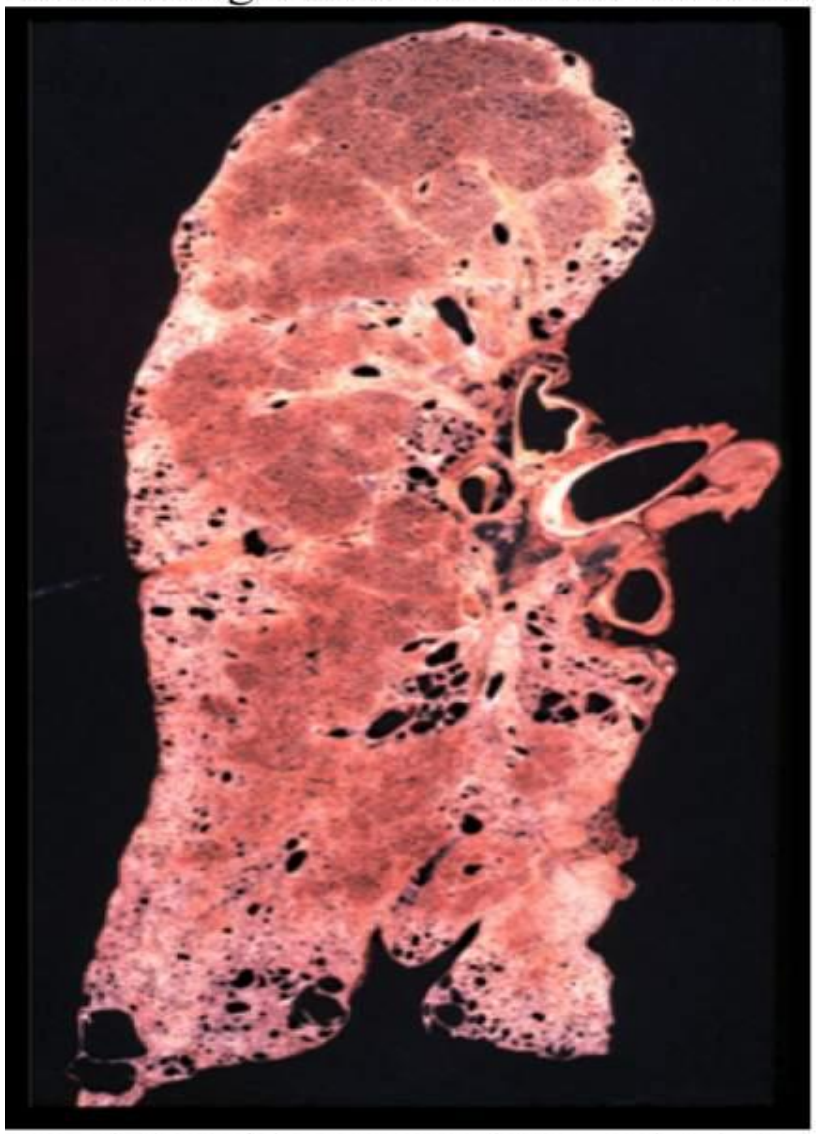
Source: Robbins and Cotran, Pathological Basis of Disease, 11th ed., 2026, Ch. 15

Asbestos body

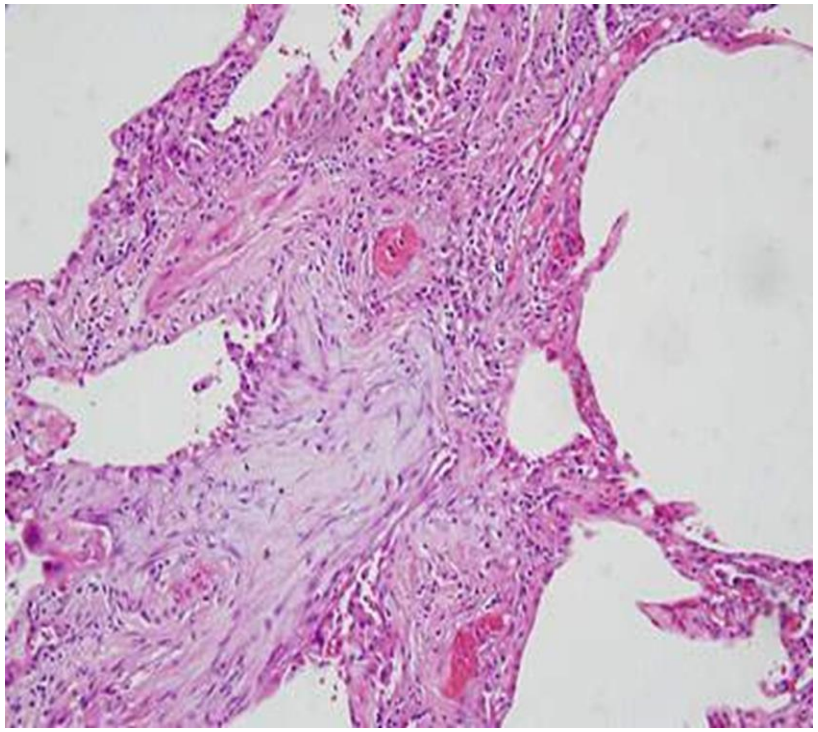


Photomicrograph shows a dumbbell-shaped asbestos body in a bronchoalveolar lavage specimen from a patient with a history of occupational asbestos exposure. Asbestos bodies differ from other nonasbestos ferruginous bodies in that the central core is thin and colorless and is covered by hemosiderin distributed in a characteristic beaded fashion. Courtesy of Jeffrey L Myers, MD.

Whole lung section of freeze-dried lung from a person who died of asbestosis



Pattern of fibrosis similar to usual interstitial fibrosis with fibroblastic foci and varying degrees of fibrosis; however, asbestos bodies will be present. Begins in lower lobes.



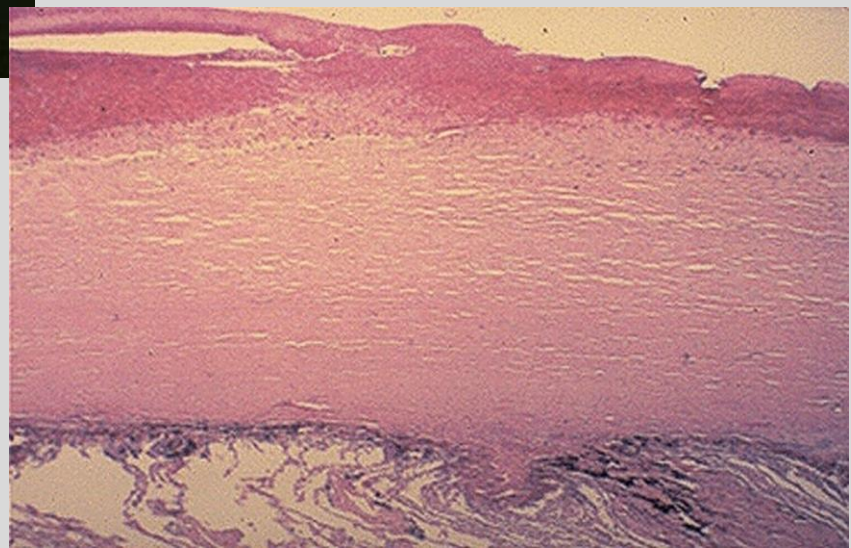
e peripheral honeycombing, which is most severe in the lower zones. Am J Respir Crit Ca
pp 692, 2004; www.atsjournals.org.

Asbestos – Further Testing



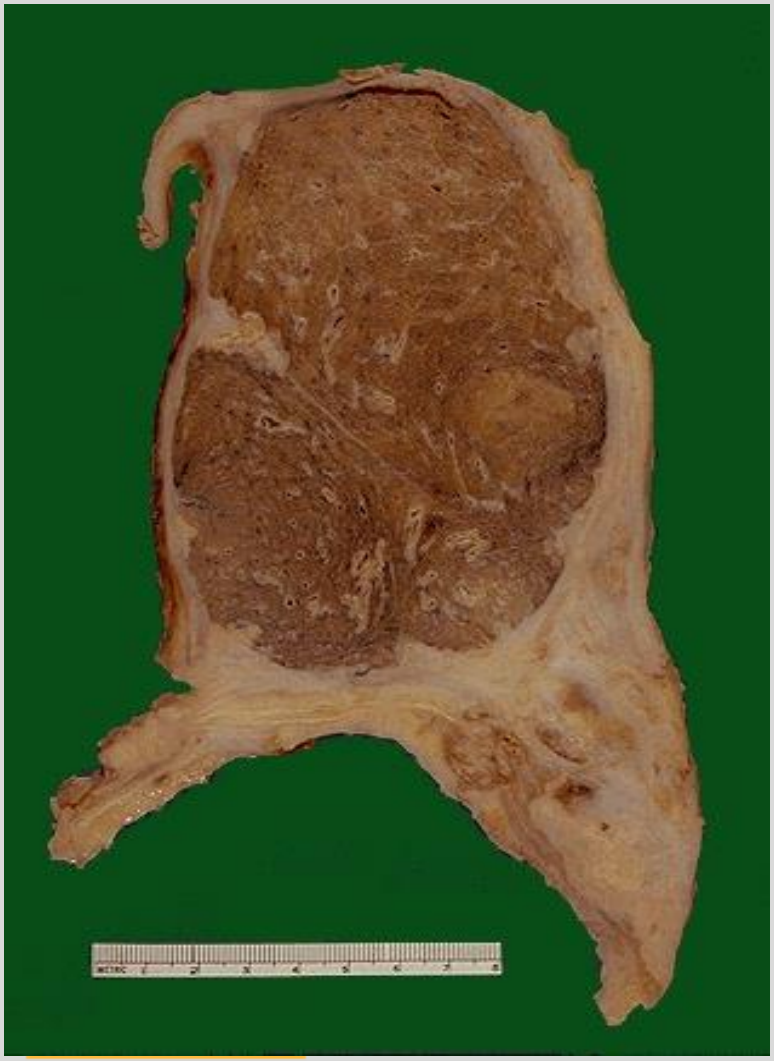
PLEURAL PLAQUES: common manifestation of asbestos exposure. Circumscribed plaques of **dense collagen, often calcified**; number and size do not correlate with level of asbestos exposure

PLEURAL PLAQUE: HISTOLOGY
dense layers of collagen; do not contain asbestos bodies, may be calcified (WebPath)



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Malignant Mesothelioma



Diffuse lesion which spreads over pleural surface and ensheaths the lung with thick tumor
Associated with pleural effusions and direct invasion of thoracic structures

https://www.studyblue.com/notes/note/n/09-lung-cancer--infxn-/deck/7661837

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Case Question

A 75-year-old man who had worked in a shipyard dies of chronic lung disease. He had also been a smoker for 40 years (1 ppd). Autopsy reveals extensive pulmonary fibrosis, and iron stains of lung tissue show numerous ferruginous bodies. The dome of the diaphragm shows the presence of pleural plaques. What is the most appropriate diagnosis?

A. Anthracosis

B. Sarcoidosis

C. Tuberculosis

D. Asbestosis

E. Silicosis

Regarding the previous question, which of the following is this person most likely to be at increased risk for?

A. Lung adenocarcinoma
B. Malignant mesothelioma
C. Miliary tuberculosis
D. Panacinar emphysema
E. Bronchiectasis

Berylliosis - Pathogenesis

- **Acute beryllium disease** in humans:
  - attributed to toxic, dose-related lung injury
  - characterized by acute inflammatory reactions in upper airways
  - bronchiolitis, pulmonary edema, and chemical pneumonitis
- **Chronic beryllium disease - multisystem** disorder characterized by noncaseating (non-necrotizing) **granulomas** that occur throughout the body; primary manifestation is in the lung, esp. affects upper lobes
- Occurs in industries such as metal machining, ceramics, and nuclear weapons manufacturing

Berylliosis – Chronic Beryllium Disease

On histopath exam, chronic beryllium disease: characterized by lymphocytic (helper/inducer T cell) alveolitis, as well as noncaseating epithelioid granulomas indistinguishable from those of sarcoidosis

Imaging: Usual finding is ill-defined nodular or irregular opacities; hilar adenopathy seen in approximately 40% of cases, but usually mild



In **later** stages of the disease, patchy fibrosis occurs with adjacent hyperinflation or distortion and extensive **honeycombing**

Clinical features of chronic beryllium disease similar to those of pulmonary sarcoidosis:

- Dyspnea, cough, chest pain, weight loss, fatigue, and arthralgias
- Physical signs may include crackles, but signs of lung disease are often absent

Berylliosis – Clinical

Pulmonary function test:

- **Usually shows a restrictive defect in advanced disease**
- **Features including obstruction**

Chronic beryllium disease

- Documented exposure to beryllium
- Evidence of lung disease compatible with the diagnosis
- Positive Beryllium lymphocyte proliferation test (BeLPT) performed on blood or BAL fluid.

http://clevelandcliniclabs.com/assets/pdfs/technical-briefs/lpt-43060.pdf

Case Question

A 69-year-old man who works in a steel yard reports to the office with a chronic cough and weight loss. On CT, irregular opacities are found, along with mild hilar adenopathy. The pulmonologist decides to biopsy the lung which reveals noncaseating epithelioid granulomas. What is the most appropriate diagnosis?

A. Silicosis

B. Chronic Obstructive Pulmonary Disease

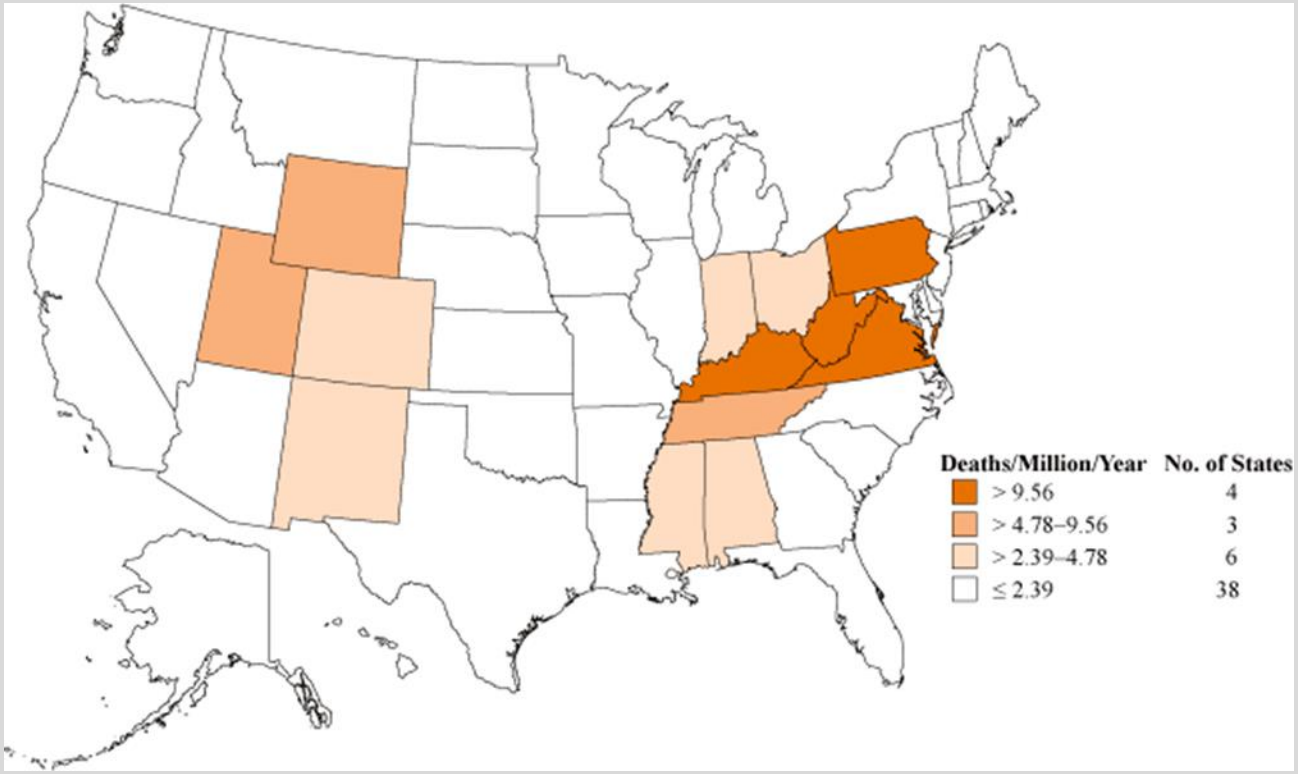
C. Berylliosis

D. Asbestosis

E. Coal Worker's Pneumoconiosis

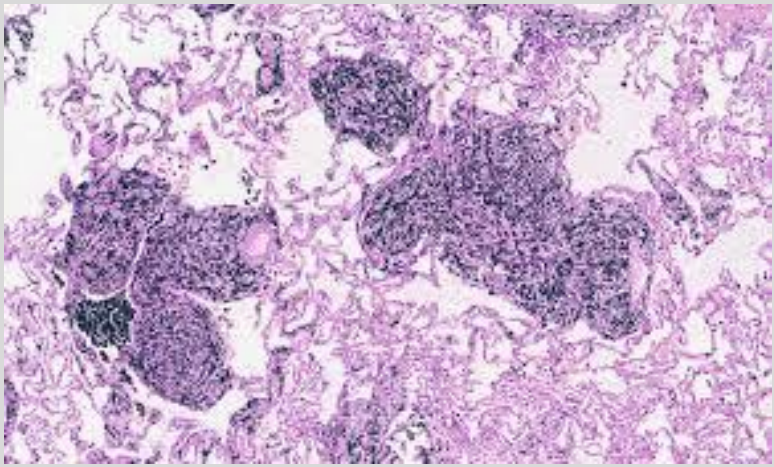
Coal Workers Pneumoconiosis

Coal workers' pneumoconiosis (CWP) is a distinct pathologic entity resulting from the deposition of **coal dust in the lungs**.



Coal Workers Pneumoconiosis

- The primary lesion in CWP is the **coal macule**
- Seen on macroscopic examination as a small (up to 4 mm) pigmented lesion
- Distributed initially in upper lobes, although lower lobes may later become involved



Coal macule (1 - 2 mm) = stellate aggregation of dust and dust-laden **macrophages**, around respiratory bronchioles, with reticulin fibers and variable amount of collagen

Coal Workers Pneumoconiosis - Pathogenesis

- Focal centriacinar emphysema, occurs within and around coal macule; together form characteristic lesion of CWP
- The **coal nodule** is a palpable lesion that, in addition to dust-laden macrophages and reticulin, contains a substantial number of **haphazardly arranged collagen fibers**
- As lung burden of dust increases, alveolar macrophages are activated and reactive oxygen species are released
- Trigger the release of cytokines, including interleukins and tumor necrosis factor, which lead to inflammation and fibrogenesis

Source: Robbins and Cotran, Pathological Basis of Disease, 11th ed., 2026, Ch. 15

Coal Workers Pneumoconiosis – Progressive Massive Fibrosis

- On background of simple CWP after many years
- Multiple, intense blackened scars **greater than 1 cm (up to 10 cm)**
- Micro shows dense collagen and pigment; center of lesion may be **necrotic because of ischemia**
- PMF develops in about 10% of cases of simple CWP
- Leads to worsening lung function, pulmonary hypertension, and cor pulmonale

Source: Robbins and Cotran, Pathological Basis of Disease, 11th ed., 2026, Ch. 15



Healthy Tissue



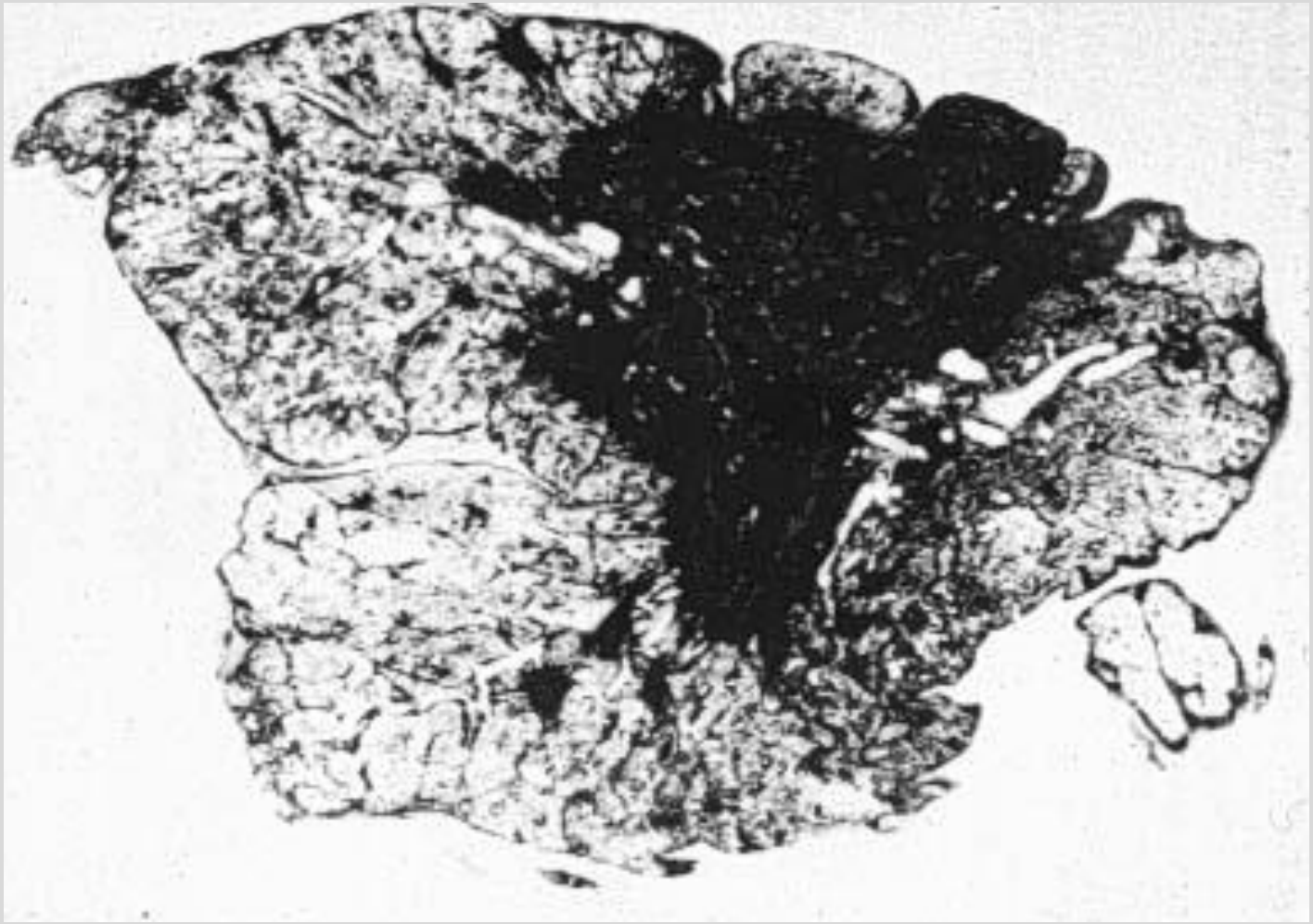
Healthy Tissue
90-year-old
schoolteacher



Progressive
massive fibrosis
40-year-old-miner

http://onradiology.blogspot.com/2010/06/some-occupational-lung-diseases-and.html

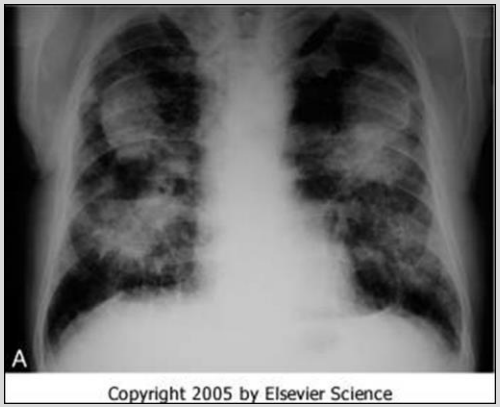
Coal Workers Pneumoconiosis – Progressive Massive Fibrosis



Coal Workers Pneumoconiosis – Progressive Massive Fibrosis

- Symptoms of cough and sputum reported by most coal miners are likely a result of dust-induced chronic bronchitis
- Breathlessness during exertion usually caused by associated chronic airflow limitation or by development of fibrosis
- Hallmark of simple CWP on the posteroanterior chest radiograph is presence of small rounded opacities in the lung parenchyma





Source: Robbins and Cotran, Pathological Basis of Disease, 11th ed., 2026, Ch. 15

Coal Workers Pneumoconiosis – Clinical

LUNG FUNCTION

Simple CWP does not really affect lung function

Coal dust has been shown to cause both **obstructive and restrictive** lung disease

Lung function deficits in complicated CWP (PMF) include:

- reduction in FVC and FEV_1
- increased total lung capacity and residual volume

MANAGEMENT AND CONTROL

Once established, the fibrotic process of CWP is thought to be irreversible. 6

Source: Robbins and Cotran, Pathological Basis of Disease, 11th ed., 2026, Ch. 15

Case Question

A 57-year-old man who lives and works in West Virginia reports to the office for a chronic wet cough. On chest X-ray, small rounded opacities in the lung parenchyma are noted. Based on his most likely diagnosis, where in the lung is this disease process most likely to first develop?

A. Right middle lobe
B. Right lower lobe
C. Left lower lobe
D. Left upper lobe
E. Bilaterally, in the lung base

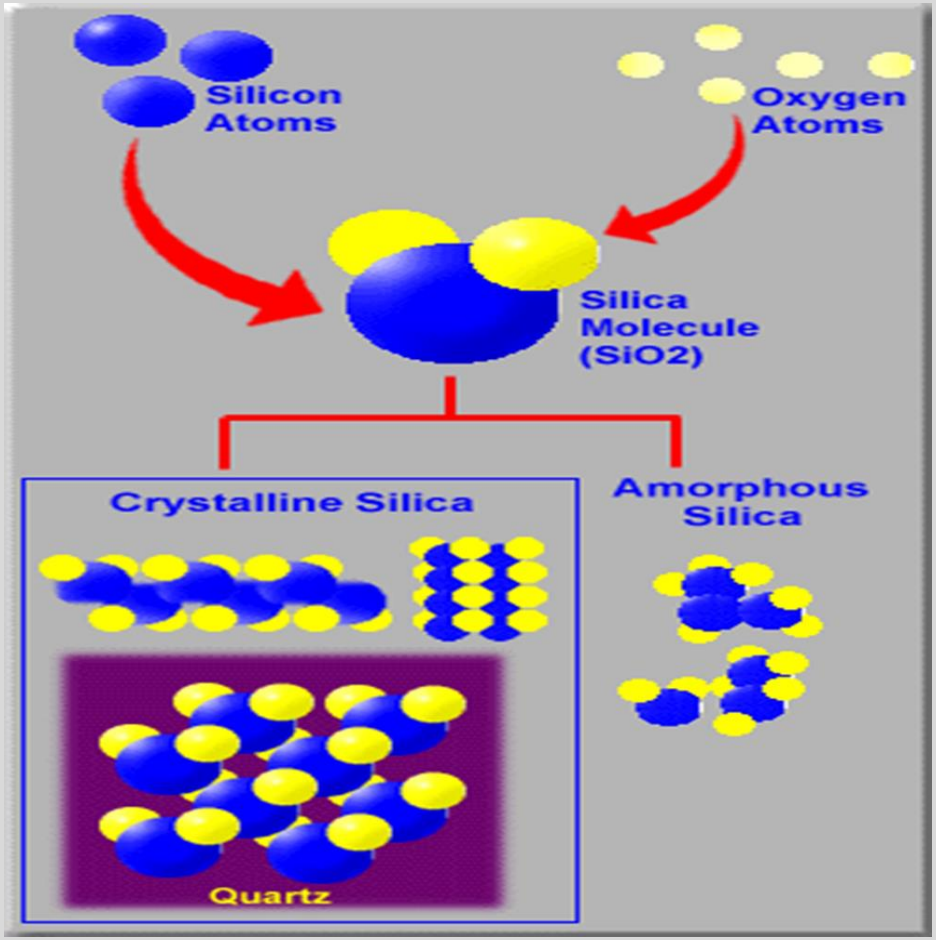
Case Question

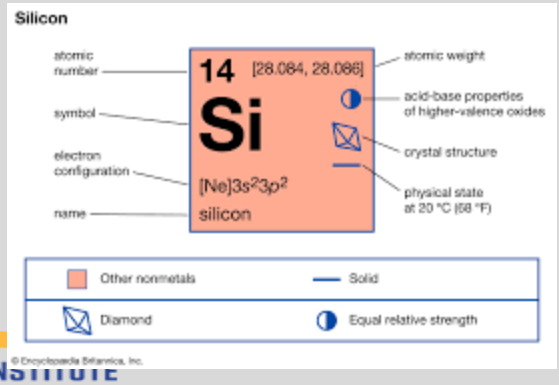
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E. Bilaterally, in the lung base

Silicosis - Background

Silicosis is a fibrotic lung disease attributable inhalation of crystalline silica, usually in the form of quartz and, less commonly, as cristobalite and tridymite





Source:Robbins and Cotran, Pathological Basis of Disease, 11th ed., 2026, Ch. 15

Waxner, Saraceno, Goblin

Silicosis - Background

1. Mining, tunneling, and excavating: miner, driller, tunneler, developer, stoper
2. Quarrying: driller, hammerer, digger
3. Stonework: cutter, dresser, driller, polisher, grinder, mason
4. Foundries: molder, knockout man, fettler, coremaker, caste
5. Ceramics: workers at any stage of process if products are dry
6. Others: glass making, boiler scaling, traditional crafts, stone grinders, gemstone workers, dental technicians, concrete reconstruction

Silicosis

Chronic silicosis, typically follows exposure, measured in decades (10 – 30 years), to respirable dust usually containing less than 30% quartz – Most common

Accelerated silicosis, follows shorter, heavier exposure (within 10 years)

Acute silicosis (silicoproteinosis), follows intense exposure to fine dust of high silica content, such as that found in **sandblasting industries**, for periods measured in weeks to months rather than years – Rare

Complicated silicosis with disease progression in both chronic and accelerated silicosis, the nodules become confluent, leading to the development of **progressive massive fibrosis (PMF)**

Source: Robbins and Cotran, Pathological Basis of Disease, 11th ed., 2026, Ch. 15
Waxner, Saraceno, Goblin

Silicosis - Pathogenesis

Chronic silicosis - most common form of the disease; most *common chronic occupational disease in world*

Hallmark of chronic silicosis **- silicotic nodule**

Mature lesion characterized by central area of whorled hyalinized collagen fibers, with a more peripheral zone of dust-laden macrophages

Silica crystals are weakly birefringent on polarization

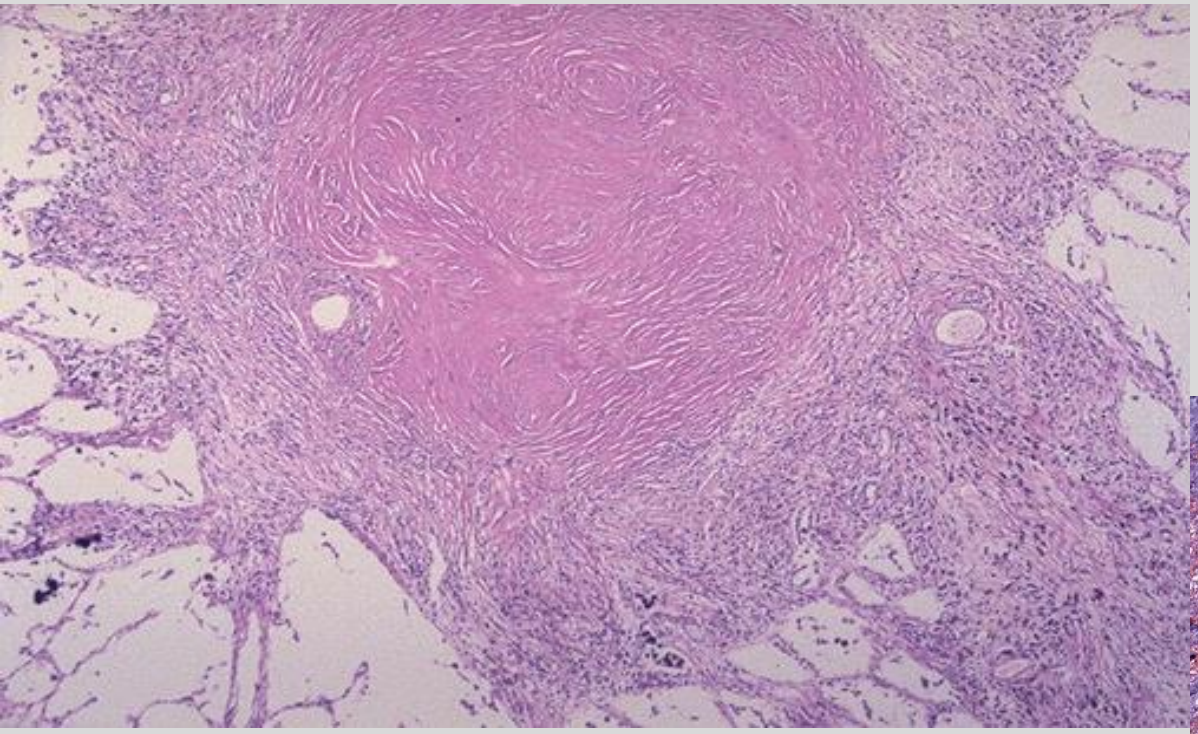
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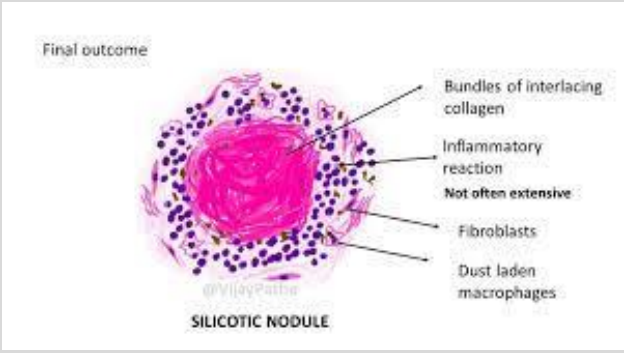
Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed., 2026, Ch. 15
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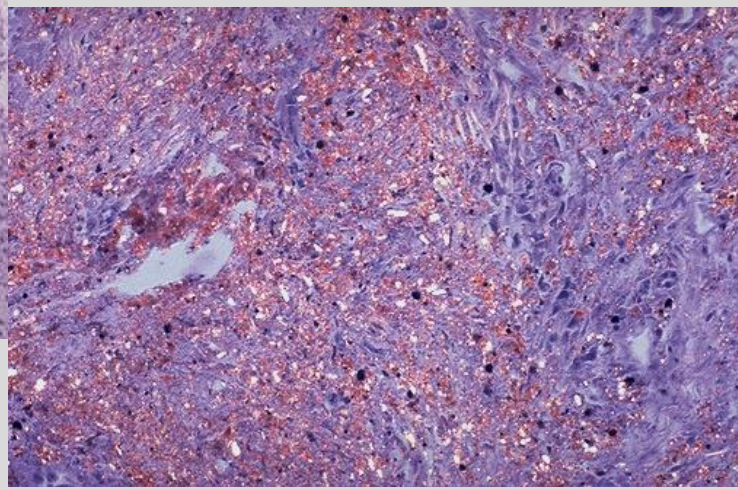
Silicosis - Pathogenesis

Nodule with concentric, interlacing layers of hyalinized collagen surrounded by dense capsule of more condensed collagen. **If polarized: birefringent silica particles**

Minimal inflammatory reaction present.







http://library.med.utah.edu/WebPath/LUNGHTML/LUNG084.html

Silicosis - Pathogenesis

- After inhalation, silica particles (crystalline are more fibrogenic than amorphous) are **phagocytosed by macrophages**
- Activates inflammasome and stimulates release of inflammatory mediators, especially IL-1 and IL-18 – leads to recruitment of additional inflammatory cells and activates interstitial fibroblasts which lay down collagen
- Freshly fractured silica, such as that generated during **sandblasting**, is more toxic to the alveolar macrophages than "aged" silica probably because of its increased redox potential

Source: Robbins and Cotran, Pathological Basis of Disease, 11th ed., 2026, Ch. 15
Waxner, Saraceno, Goblin

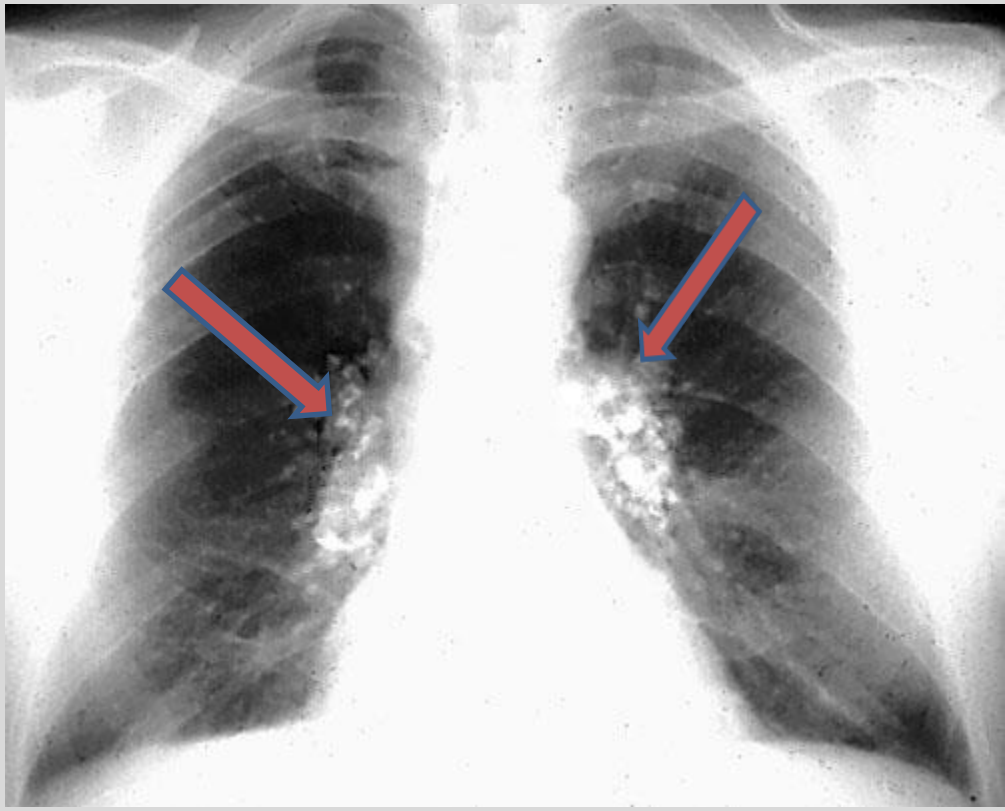


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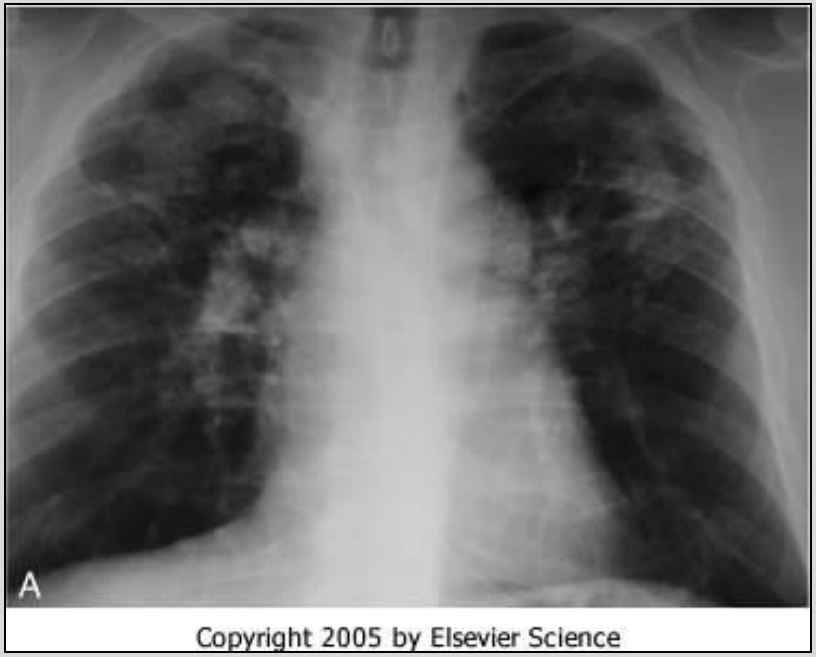
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Hard, collagenous scar of silicosis (blackened)
Dense pleural thickening is also present. (photo, left)

Thin rim of calcification around hilar lymph nodes may be seen on Xray **(eggshell calcification)** - below



Patient with chronic silicosis *may present without symptoms* for assessment of an abnormal chest radiograph



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Posteroanterior radiograph showing features of progressive massive fibrosis (PMF) **with "eggshell" calcification of the hilar nodes**, in keeping with complicated silicosis, in a man who had shoveled sand in a glass factory

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Downloaded from: Textbook of Respiratory Medicine (on 20 October 2008 12:49 AM)

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Silicosis & Tuberculosis

- Association between silicosis and tuberculosis has long been recognized
- May be related to the fact that crystalline **silica inhibits ability of pulmonary macrophages to kill phagocytosed mycobacteria**
- Rates for active tuberculosis in silicotic patients range from 2- to 30-fold more than those in the same workforce without silicosis

Robbins and Cotran, Pathologic Basis of Disease, 11th ed., 2026, Ch. 15
Waxner, Saraceno, Golbin

Silicosis - Clinical

LUNG FUNCTION

- PFTs may be normal but may become abnormal if progressive massive fibrosis develops
- Spirometric tests (FEV_1, FEV_1/FVC, and maximal midexpiratory flow) usually reflect airflow limitation
- Reduction in diffusing capacity is generally apparent in more advanced chronic silicosis

MANAGEMENT AND CONTROL

Once established, the fibrotic process of chronic silicosis is thought to be irreversible

Case question

A 60-year-old mason complains of shortness of breath, which has become progressively worse during the past year. A chest Xray shows small nodular shadows in both lungs. PFTs reveal a pattern consistent with restrictive lung disease. The patient subsequently develops congestive heart failure and expires. Autopsy discloses numerous small fibrotic nodules in both lungs. Histological examination of these nodules show concentric, whorling, interlacing dense, hyalinized collagen fibers with no granulomas. What is the most likely diagnosis?

A. Tuberculosis

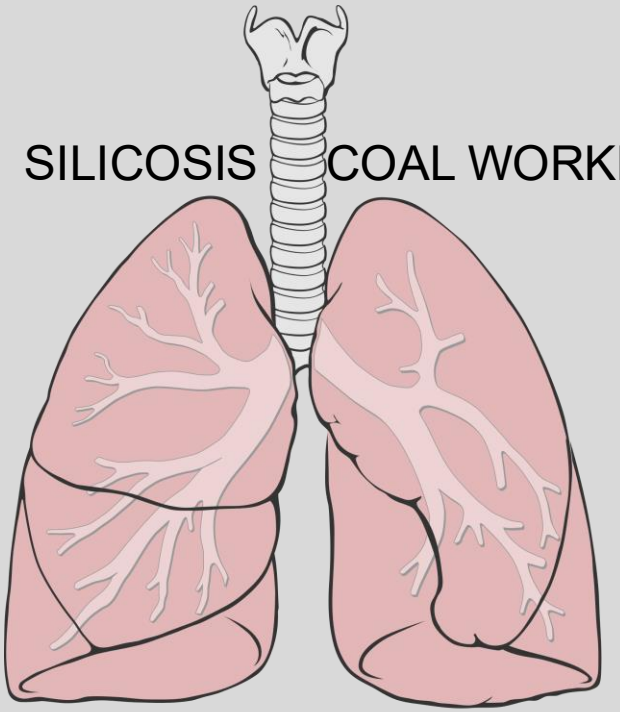
B. Asbestosis

C. Anthracosis

D. Silicosis

E. Sarcoidosis

Pneumoconioses	Asbestos is from the roof (was common in insulation), but affects the base (lower lobes). Silica and coal are from the base (earth), but affect the roof (upper lobes).



SILICOSIS COAL WORKER'S PNEUMOCONIOSIS

ASBESTOSIS

Source: First Aid (2020)

Summary Slide

- Definition of Pneumoconiosis – restrictive lung disease/occupations
- Asbestos-related disease – pathogenesis, clinical, lab, Xray
  - Asbestos fibers
  - Pleural plaques and effusions
  - Diffuse interstitial lung fibrosis
  - Malignant mesothelioma
- Berylliosis – pathological and immune pathogenesis
  - 17% of acute beryllium disease progress to chronic
  - Non-caseating granulomas, interstitial lung disease
  - Clinical similar to sarcoidosis
- Coal Workers' Pneumoconiosis
  - Coal macule and coal nodule (coal macule – macro/micro: aggregate of dust laden macrophages around resp. bronchiole with reticulin fibers and collagen)
  - Complicated CWP – progressive massive fibrosis – on background of simple CWP after many years
- Silicosis – chronic, accelerated, acute
  - Silicotic nodule
  - Pathogenesis
  - Associated with TB and lung cancer

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Lecture Feedback Form:

https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R